MIL-S-1491D
19 August 1977
SUPERSEDING
MIL-S-1491C
27 March 1970

MILITARY SPECIFICATION

SOAPBOX, PLASTIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of plastic soapbox.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-390	-	Plastic, Molding Material, Polyethylene, and Copolymers (Low, Medium, and High Density)
P-S-620	-	Soap, Toilet, (Cake and Powdered)
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8530

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Plastic molding compound. The plastic molding compound shall be virgin polyethylene conforming to type I, class L, grade 1 or 2 of L-P-390. Clean, unburned, reground material from imperfect parts, sprues, and runners due to molding of the item and of the same composition as the virgin material, may be reused by adding to the virgin material. Not more than 15 percent clean, unburned reground material specified above shall be permitted in any one batch.

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3.2 Construction. The plastic soapbox shall be rectangular in shape, consisting of two parts, a tray and a cover. These parts shall be of single piece construction with all corners well rounded. The two parts of the soapbox shall fit snugly together when telescoped. Semi-circular finger grips shall be centrally located on the bottom edges of the long sides of the cover. Small stops or shoulders shall be located in each inside corner, or four stops spaced along the inside of the cover to control the depth of fit. The stops or shoulders shall be not less than 5/8 inch (16 mm) from the bottom edges of the cover. The outside top of the cover shall be stiffened by beading in both the longitudinal and crosswise direction. The inside bottom of the tray shall be beaded or corrugated in such a way that the bar of soap is not completely in contact with the plastic material.

3.2.1 Dimensions. The dimensions of the plastic soapbox shall be as specified in table I.

TABLE I. Dimensions

		Minimum Inches (mm)	Maximum Inches (mm)
Tray	Length (inside)	3-1/4 (83)	3-5/8 (92)
	Width (inside)	2-3/16 (56)	2-7/16 (62)
	Depth (inside)	3/4 (19)	1 (25)
	Wall thickness	5/64 (2)	-
Cover	Length (inside) 1/		
	Width (inside) 1/		
	Wall thickness	5/64 (2)	-
Assembled box	Overall height (outside)	1-1/4 (32)	1-1/2 (38)

1/ Length and width of cover to provide telescoping, snug fit with tray.

3.3 Performance.

3.3.1 Warpage. The soapbox shall show no warpage or distortion when tested as specified in 4.4.

3.3.2 Impact strength. The plastic soapbox shall show no evidence of cracking or breaking when tested as specified in 4.4.

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3.3.3 Stress resistance. The cover and tray of the plastic soapbox shall show no cracking or separation when tested as specified in 4.4.

3.4 Color. The color of the finished soapbox shall approximate color No. 14087 of FED-STD-595.

3.5 Marking for identification. Each plastic soapbox shall be permanently and distinctly marked, by an embossed molded in process, with the name or brand name of the manufacturer, of such known character as to be easily identified. The manufacturer's name or brand name shall be located on the outside of the bottom of the tray.

3.6 Workmanship. The soapbox shall be uniformly colored and free from any discoloration, bloom, burn, scale, faded or chalked area. The surface of the box shall be clean and smooth with a uniform texture free from any thin spot, wrinkle, pimple, crater, dome, dish or sink mark, malformation, pit, porous area, bubble (open or cloud), heat mark, cold seam, blister, rough surface, crack, break, rupture, fracture, scratch, dent, nick, oil, dirt or foreign matter. The edges of the box shall be smooth and free from any fine, flash, rough or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

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4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for defects set forth in the applicable subparagraphs, at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. The lot size, for purposes of determining the sample size, shall be expressed in units of soapboxes for examination in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examination in 4.3.1.3.

4.3.1.1 Visual examination. The sample unit for this examination shall be one soapbox. The soapbox shall be examined for defects in material, design, construction, appearance and workmanship.

Examine	Defect
Color	Not as specified. Discolored, faded or mottled; not uniform in color, chalked, burned bloomed or scaly.
Construction and workmanship	Not free of rough or sharp edges. Not rectangular. Not smooth and uniform in texture; thin spot, wrinkle, pimples or craters, short, pulled or rough surface; domed, dished or sink marked; malformed, crazing, pitted, porous or bubble (open or closed); heat mark, cold seam, or blister. Tray not single piece construction. Cover not single piece construction. Tray and cover do not fit together snugly when telescoped. Semi-circular finger grips missing. No stops or shoulder on each inside corner or along the sides of the cover. Bottom of tray not beaded or corrugated. Top of cover not beaded in both longitudinal and crosswise direction. Any crack, break, rupture or fracture. Oil, dirt, or foreign matter. Any scratch, dent or nick on outside of box affecting appearance. Inadequate number of stops or shoulders on inside corners or spaced along the inside of the cover.
Marking for identification	Manufacturer's name or brand name missing from outside bottom of tray.

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4.3.1.2 Dimensional examination. The sample unit for this examination shall be one soapbox. The soapbox shall be examined for defects in dimensions. Any dimensions that do not conform to the dimensional requirements specified in table I shall be classified a defect.

4.3.1.3 Packaging inspection. An examination shall be made to determine that preservation-packaging, packing and marking comply with the requirements of section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it shall not be closed. Examination for the closure defects listed below shall be made on shipping containers fully prepared for delivery. The lots size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location; sequence or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components; such as incomplete closure of container flap, improper taping, loose or inadequate sealing, strapping or stapling. Bulged or distorted container.
Contents	Number of soapboxes per unit package is more or less than required. Number of unit packages per shipping container is more or less than required.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs). The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per one hundred units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-1	2.5
4.3.1.3	S-2	2.5

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4.3.2 Testing of the end item. Testing of the completely fabricated plastic soapbox shall be performed in accordance with 4.4 for the characteristics shown therein. The sample unit shall be two finished soapboxes. The inspection level shall be S-1 of MIL-STD-105. The lot size shall be expressed in units of one soapbox. The lot shall be unacceptable if any sample unit fails to meet any requirement specified.

4.4 Test procedures.

4.4.1 Conditioning. The warpage, impact strength, and stress resistance tests shall be performed at standard laboratory conditions 73.5 ± 2°F (23 ± 1.1°C) and 50 ± 4 percent relative humidity.

4.4.2 Warpage test. A three percent aqueous solution of any soap, of any type conforming to P-S-620, shall be placed in the bottom section of the specimen soapbox in sufficient quantity to completely cover the bottom surface. The soapbox shall then be assembled and placed in an oven at 120°F (49°C) for 24 hours. After returning to room temperature, the soap shall be washed out and the soapbox allowed to dry. The soapbox shall be examined to determine compliance with 3.3.1.

4.4.3 Impact strength. Both the cover and the tray sections of the specimen soapbox are to be tested after the warpage test has been performed. The section under test shall be placed open side down on a wooden block having a smooth surface. The block shall be supported by a nonresilient surface. A steel ball, approximately 2 inches (51 mm) diameter, shall be dropped from such a height as to generate an impact equal to one foot-pound (1.35 J). The steel ball dropped twice on the center of the soapbox section shall constitute the test. The soapbox shall be examined to determine compliance with 3.3.2.

4.4.4 Stress resistance. All four sides of the over section shall be individually flexed by bending inward with the thumb until contact is made with the top. The thumb pressure shall be applied at approximately the center of each side. The same test shall be conducted on the tray section of the soapbox. The soapbox shall be examined to determine compliance with 3.3.3.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. Twelve assembled soapboxes shall be preserved-packaged in a folding paperboard box conforming to variety 1, style III, type C, class 1, group I of PPP-B-566; or setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. The soapboxes shall be arranged on end, three in length, four in width and one in depth within a paperboard box. Inside dimensions of each interior box shall be 6-3/4 inches in length (171 mm), 6 inches (152 mm) in width and 3-7/8 inches (98 mm) in depth. The box closure shall be secured with 2 inch (51 mm) minimum width gummed paper tape conforming to type III, grade B of PPP-T-45; applied at the center of the length opening and extending along the bottom and up each side at least 2 inches (51 mm).

5.1.2 Level C. Soapboxes shall be preserved-packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A, packing. Seven hundred and twenty soapboxes, preserved-packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A packages shall be packed flat, four in length, three in width and five in depth within a shipping container. Inside dimensions of each shipping container shall approximate 25-1/2 inches (648 mm) in length, 21-1/2 inches (546 mm) in width and 20 inches (508 mm) in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.1.3.

5.2.2 Level B, packing. Seven hundred and twenty soapboxes, preserved-packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF, (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A packages shall be packed flat, four in length, three in width and five in depth within the shipping container. Inside dimensions of each shipping container shall approximate 25-1/2 inches (648 mm) in length, 21-1/2 inches (546 mm) in width and 20 inches (508 mm) in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.3.1.3.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.3.1.3.

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5.2.3 Level C, packing. Soapboxes, preserved-packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract, interior packages, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The plastic soapbox is intended for use as a personal toilet soap container by military personnel.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 99

User activities:

Army - MD
Navy - MC

Preparing activity:

Army - GL

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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